Enrollment No : 240840116021

Practical 7

AIM : Develop advanced visualizations (Box Plot, Heatmap, Pair Plot, Violin Plot).

Task 1 : Create and Inspect the dataset

In [1]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

In [2]:
```
# create the dataset
data = {
    "Student_Id": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14],
    "Study_Hours": [1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 8.0, 9.0],
    "Attendance": [65, 70, 72, 75, 78, 80, 82, 85, 87, 90, 91, 93, 95, 97],
    "Assignment_Score": [55, 60, 62, 65, 68, 72, 75, 78, 80, 84, 87, 90, 94, 98],
    "Midterm_Marks": [48, 52, 55, 58, 61, 65, 68, 72, 75, 79, 82, 86, 91, 96],
    "Final_Marks": [45, 50, 52, 57, 62, 68, 72, 76, 81, 85, 89, 93, 97, 100]
}
```

In [3]:
```
# create the pandas data frame
df = pd.DataFrame(data)
```

In [4]:
```
# disoplay the first five records of the dataframe
df.head()
```

Out[4]:

	Student_Id	Study_Hours	Attendance	Assignment_Score	Midterm_Marks	Final_Marks
0	1	1.5	65	55	48	45
1	2	2.0	70	60	52	50
2	3	2.5	72	62	55	52
3	4	3.0	75	65	58	57
4	5	3.5	78	68	61	62

In [5]:
```
# display information about columns and data types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Student_Id        14 non-null     int64
 1   Study_Hours       14 non-null     float64
 2   Attendance        14 non-null     int64
 3   Assignment_Score  14 non-null     int64
 4   Midterm_Marks     14 non-null     int64
 5   Final_Marks       14 non-null     int64
dtypes: float64(1), int64(5)
memory usage: 804.0 bytes
```

In [6]:
```
# display basic statistical information using describe
df.describe()
```

Out[6]:

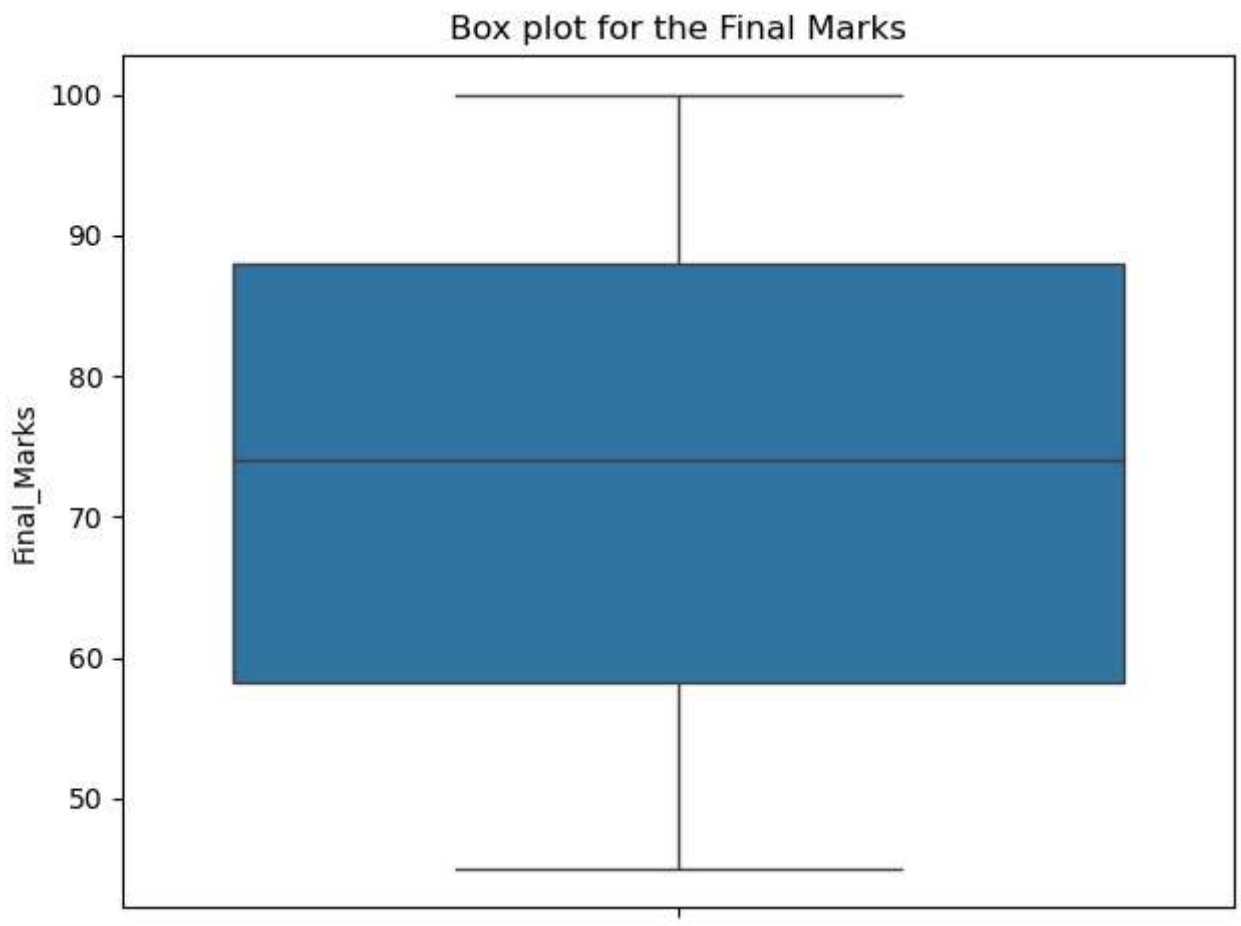
	Student_Id	Study_Hours	Attendance	Assignment_Score	Midterm_Marks	Final_Marks
count	14.0000	14.000000	14.000000	14.000000	14.000000	14.000000
mean	7.5000	4.857143	82.857143	76.285714	70.571429	73.357143
std	4.1833	2.273836	9.944903	13.309503	14.960020	18.257168
min	1.0000	1.500000	65.000000	55.000000	48.000000	45.000000
25%	4.2500	3.125000	75.750000	65.750000	58.750000	58.250000
50%	7.5000	4.750000	83.500000	76.500000	70.000000	74.000000
75%	10.7500	6.375000	90.750000	86.250000	81.250000	88.000000
max	14.0000	9.000000	97.000000	98.000000	96.000000	100.000000

Task 2 : Create a Box Plot

In [7]:
```
# create the box plot to check the data spreadness
sns.boxplot(
    data = df,
    y = "Final_Marks"
```

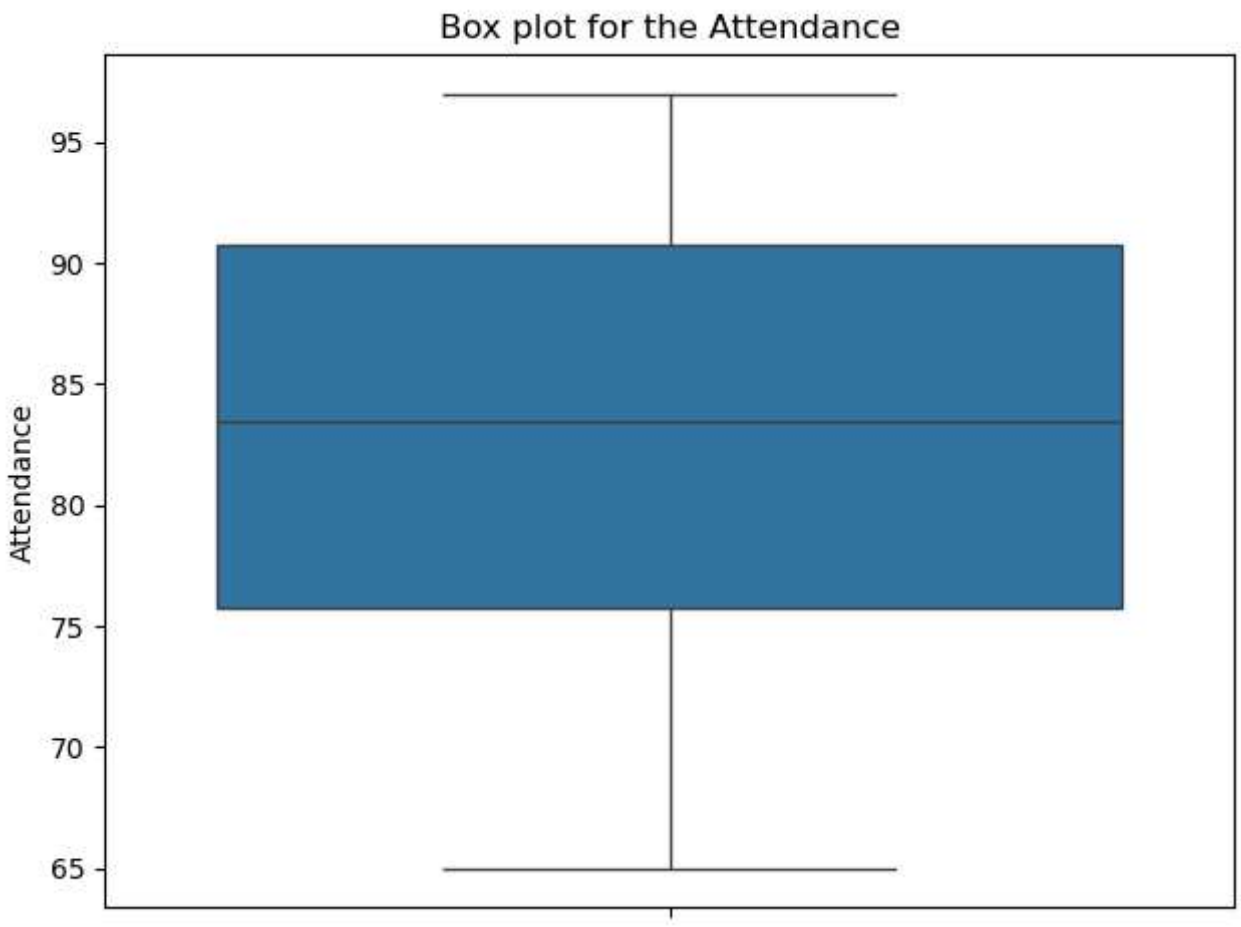
```
)

plt.title("Box plot for the Final Marks")
plt.tight_layout()
plt.show()
```



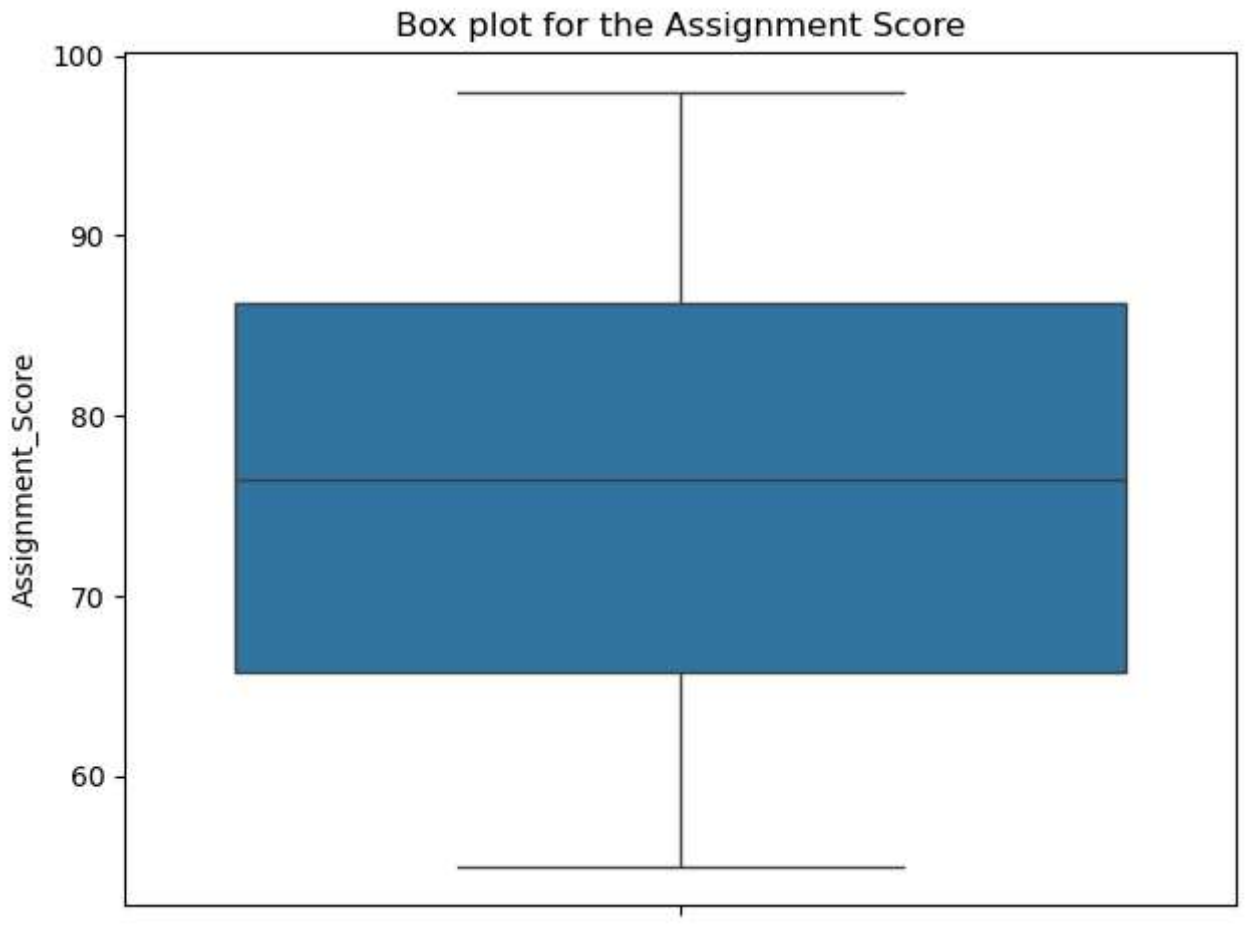
In [8]:
```
# box plot for the attendance
sns.boxplot(
    data = df,
    y = "Attendance"
)

plt.title("Box plot for the Attendance")
plt.tight_layout()
plt.show()
```



In [9]:
```
# box plot for the assignment score
sns.boxplot(
    data = df,
    y = "Assignment_Score"
)

plt.title("Box plot for the Assignment Score")
plt.tight_layout()
plt.show()
```



Task 3 : Create a Heatmap

In [10]:
```
# extract the numerical columns form the dataframe
numerical_cols = ["Study_Hours" ,"Attendance"   ,"Assignment_Score"     ,"Midterm_Marks"        ,"Final_Marks"]

# find the correlation between columns inside the data frame
corr_matrix = df[numerical_cols].corr()
```

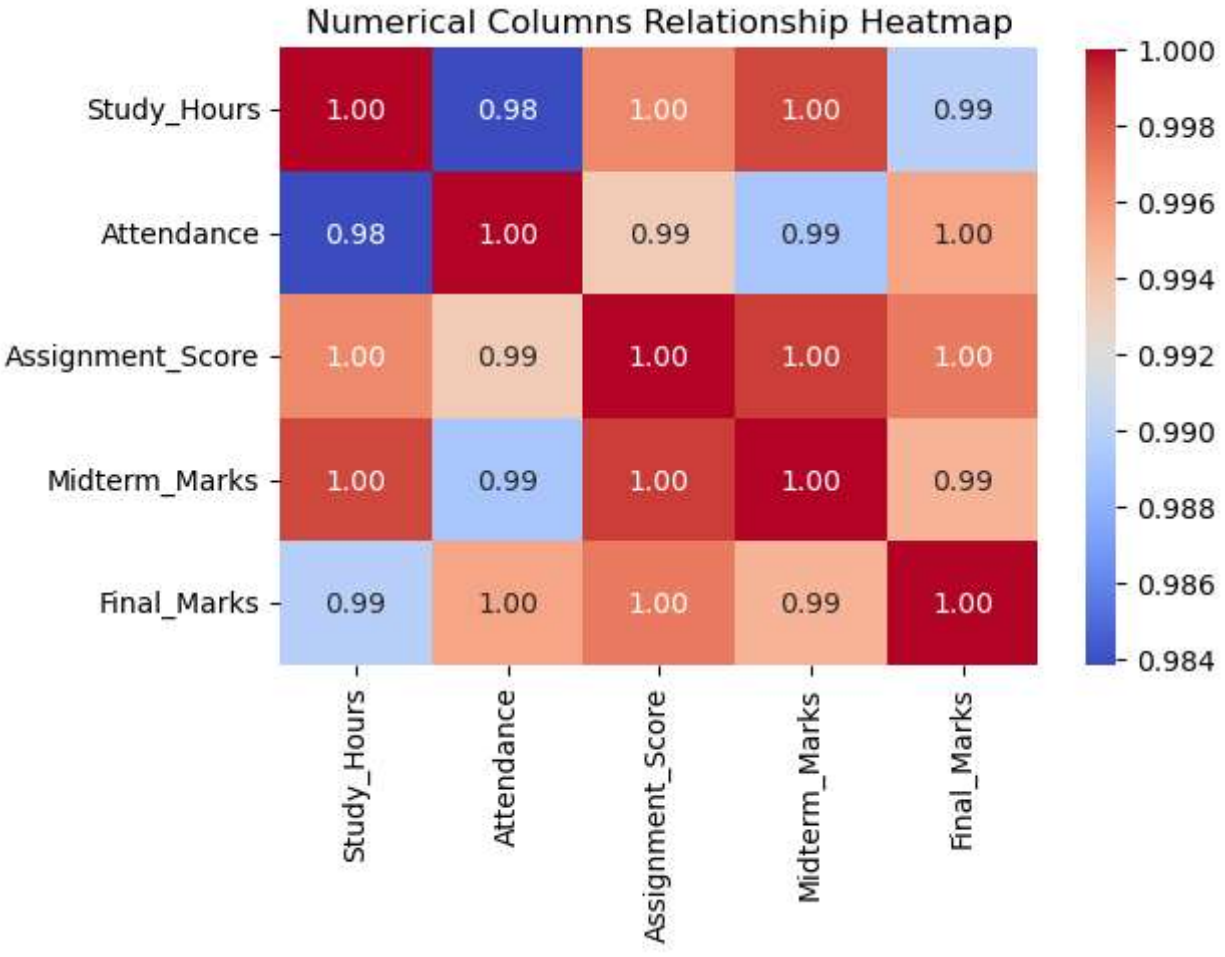
In [11]:
```
# dispaly the correlation matrix
print(corr_matrix)
```

```
                 Study_Hours  Attendance  Assignment_Score  Midterm_Marks  \
Study_Hours         1.000000    0.983822          0.996554       0.998703
Attendance          0.983822    1.000000          0.993531       0.989171
Assignment_Score    0.996554    0.993531          1.000000       0.998949
Midterm_Marks       0.998703    0.989171          0.998949       1.000000
Final_Marks         0.989873    0.995492          0.997041       0.994785

                 Final_Marks
Study_Hours         0.989873
Attendance          0.995492
Assignment_Score    0.997041
Midterm_Marks       0.994785
Final_Marks         1.000000
```

In [12]:
```
# create the heatmap for this numerical columns
sns.heatmap(
    corr_matrix,
    cmap = "coolwarm",
    annot = True,
    fmt = ".2f"
)

plt.tight_layout()
plt.title("Numerical Columns Relationship Heatmap")
plt.show()
```



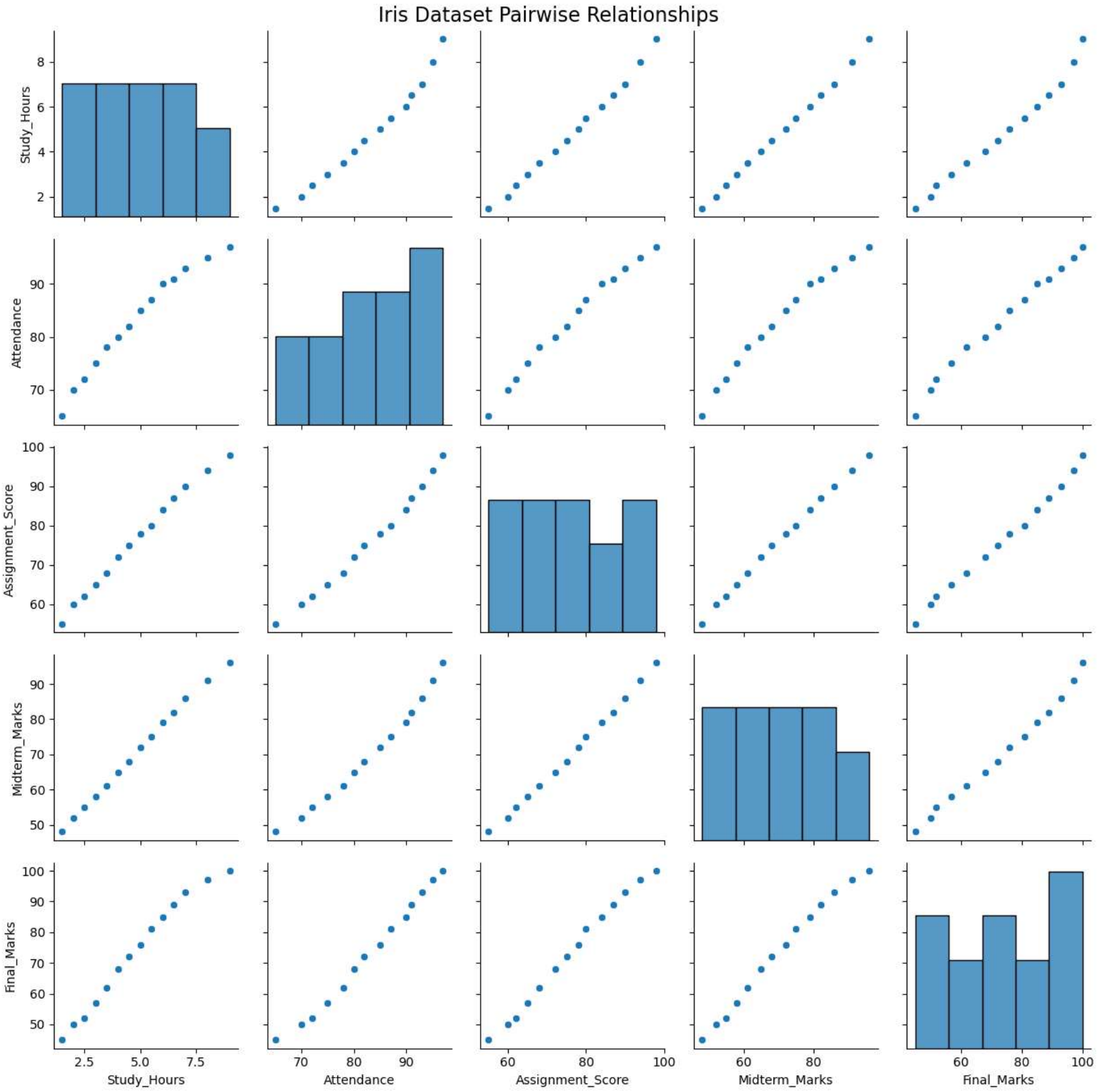
Task 4 : Create a Pair Plot

In [13]:
```
# create the numerical values pair plot for the trend identification


sns.pairplot(
    data = df[numerical_cols]
)

# add the title of the pair plot explicitly
plt.suptitle("Iris Dataset Pairwise Relationships", fontsize=16)
plt.subplots_adjust(top=0.95)

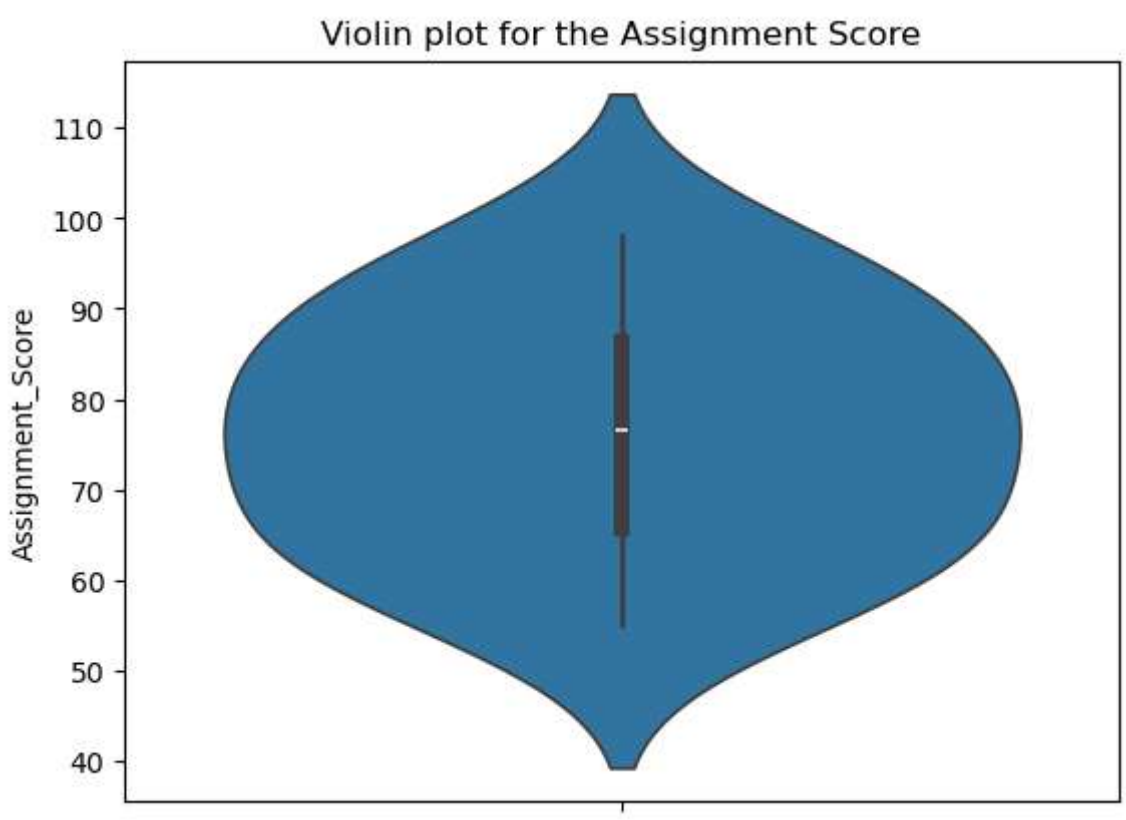
plt.tight_layout()
plt.show()
```



Task 5 : Create a Violin Plot

In [14]:
```
# violin plot for the Assignment_Score
sns.violinplot(
    data = df,
    y = "Assignment_Score"
)

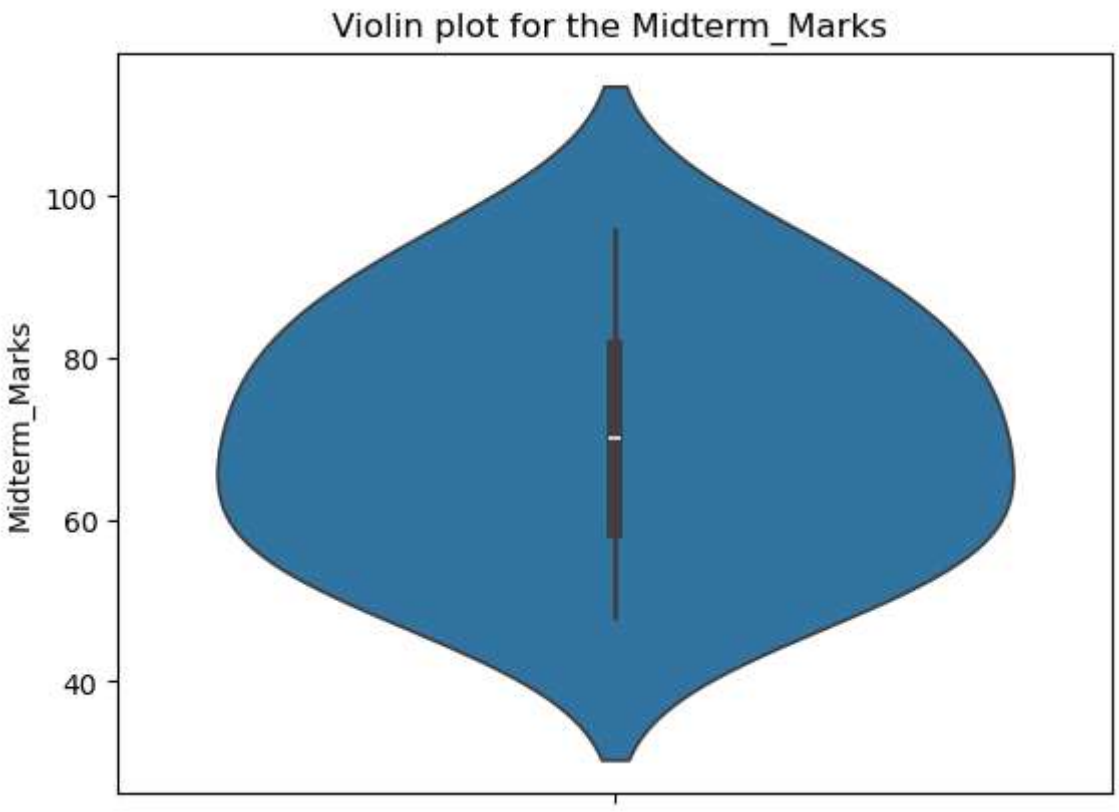
plt.title("Violin plot for the Assignment Score")
plt.show()
```



In [15]:
```
# violin plot for the Midterm_Marks
sns.violinplot(
```

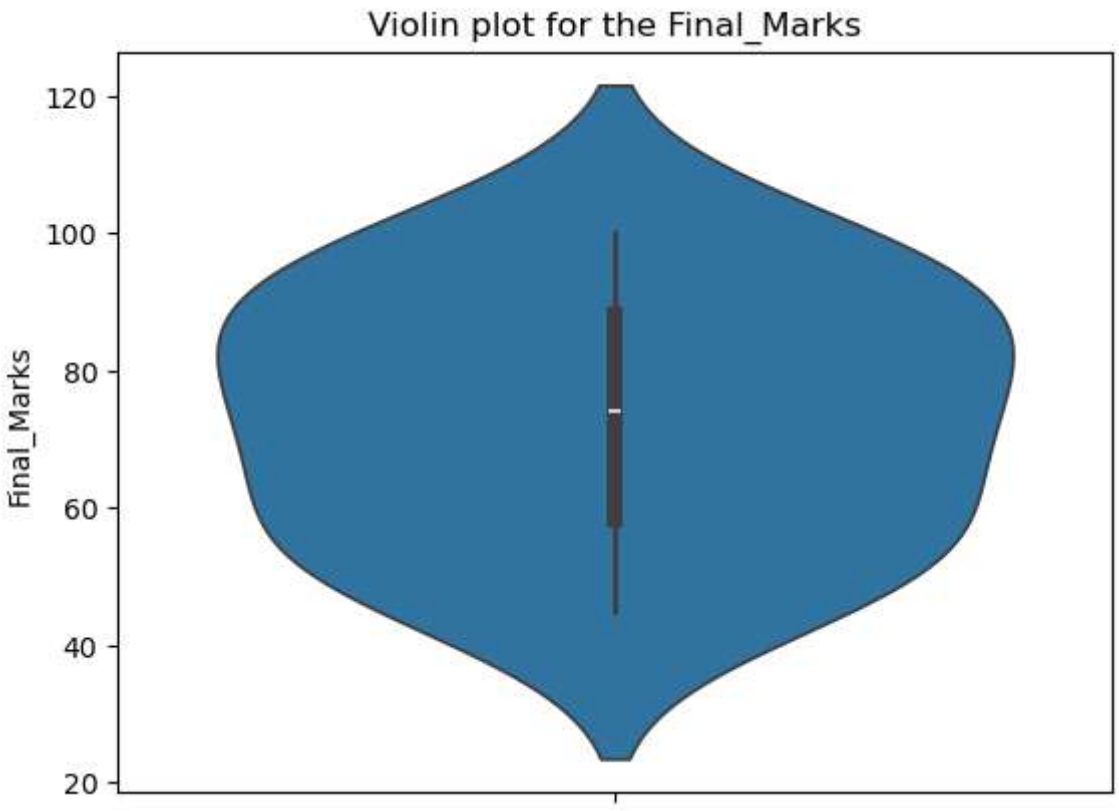
```
    data = df,
    y = "Midterm_Marks"
)

plt.title("Violin plot for the Midterm_Marks ")
plt.show()
```



In [16]:
```
# violin plot for the Final_Marks
sns.violinplot(
    data = df,
    y = "Final_Marks"
)

plt.title("Violin plot for the Final_Marks")
plt.show()
```



In [17]:
```
# violin plots for teh Assignment_Marks, Midterm_Marks, Final_Marks
fig, axes = plt.subplots(1, 3, figsize=(12, 5))

sns.violinplot(
    ax = axes[0],
    data = df,
    y = "Assignment_Score"
)

axes[0].set_title("Violin plot for the Assignment Score")

# violin plot for the Midterm_Marks
sns.violinplot(
    ax = axes[1],
    data = df,
    y = "Midterm_Marks"
)

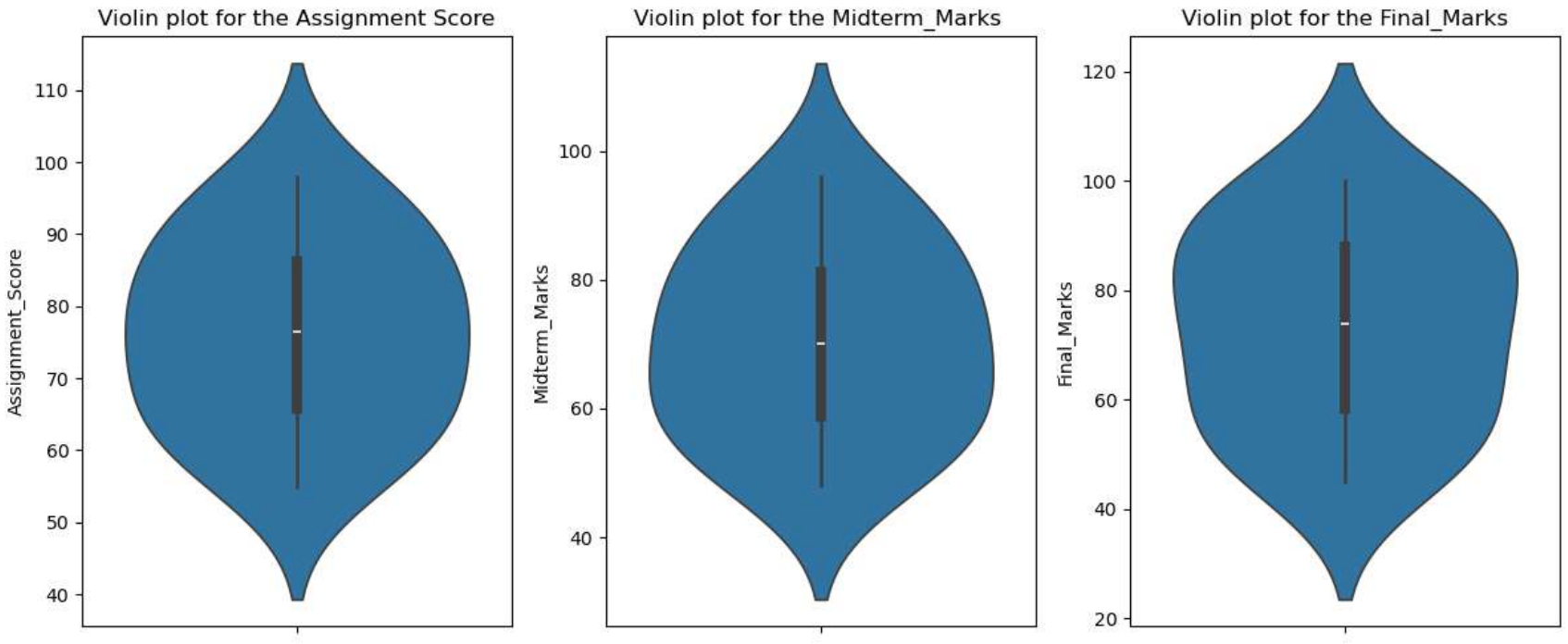
axes[1].set_title("Violin plot for the Midterm_Marks ")

# violin plot for the Final_Marks
sns.violinplot(
    ax = axes[2],
    data = df,
```

```
    y = "Final_Marks"
)

axes[2].set_title("Violin plot for the Final_Marks")

plt.tight_layout()
plt.show()
```



Task 6 : Save the Dataset

In [18]:
```
# save the dataset
df.to_csv("student_dataset.csv")
print("Dataset saved sucessfully...!!!")
```

```
Dataset saved sucessfully...!!!
```

In [20]:
```
# download csv from the google collab
from google.colab import files

# 'your_file.name.csv' ki jagah apni file ka exact naam likhein
files.download('student_dataset.csv')
```